CineMatch

Movie Recommendation System

Low-Level Design (LLD) Document

API Endpoint Definitions & I/O Specifications | Version 1.0 | April 2026

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1. API Overview

Base URL (local)	http://localhost:8000
Base URL (Docker)	http://backend:8000
Content-Type	application/json
Authentication	None (TMDB API key stored server-side in .env)
Framework	FastAPI 0.110.0
API Docs	http://localhost:8000/docs (Swagger UI auto-generated)
CORS	Allow all origins in development

2. Endpoint Definitions

2.1 GET /

Root health check. Confirms API is running.

Method	GET
URL	/
Parameters	None
Auth	None

Response (200):

```
{ "message": "CineMatch API is running!" }
```

2.2 GET /health

Health check. Returns uptime and version. Used by Docker healthcheck and orchestration.

Method	GET
URL	/health
Parameters	None

Response (200):

```
{ "status": "healthy", "uptime": "0:12:34", "version": "1.0.0" }
```

2.3 GET /ready

Readiness check. Verified model pkl files exist and TMDB API key is set.

Method	GET
URL	/ready
Parameters	None

Response (200):

```
{ "status": "ready", "model_version": "20260412_154312", "api_key_set": true }
```

Response (503) — not ready:

```
{ "detail": { "status": "not ready", "issues": ["Missing model file: movies.pkl"] } }
```

2.4 GET /recommend [PRIMARY ENDPOINT]

Returns 10 movie recommendations for a query. Supports movie title, genre, actor name, and character name search.

Method	GET
URL	/recommend
Parameter	movie: string (required) — movie title, genre, actor, or character name
Fuzzy Matching	Cutoff 0.5 — tolerates minor typos
Sort Order	1. Searched movie first 2. Highest IMDB rating 3. No rating (N/A)

Example request:

```
GET /recommend?movie=Inception
```

Example response (200):

```
{ "recommendations": [
    { "title": "Inception", "poster": "https://image.tmdb.org/t/p/w500/...",
      "rating": 8.4, "overview": "A thief who steals corporate secrets..." },
    { "title": "Interstellar", "poster": "https://...", "rating": 8.6, "overview":
"..." }
  ] }
```

HTTP Code	Description
200	Recommendations returned successfully
400	Movie name is empty or missing
404	Movie not found in dataset after fuzzy matching
500	Internal server error (model or TMDB failure)

Search type detection order:

- 1. Exact genre match (action, romance, horror, sci-fi...) → genre search
- 2. Character name match (loki, batman, hermione...) → actor search
- 3. Actor pre-check (3+ movies found in tags) → actor search
- 4. Fuzzy title match (cutoff 0.5) → similarity search
- 5. Fallback → actor/character search

2.5 GET /metrics

Prometheus metrics scrape endpoint. Returns all instrumented metrics in text format.

Method	GET
URL	/metrics
Scraped by	Prometheus every 10 seconds
Content-Type	text/plain; version=0.0.4

Sample output:

```
cinematch_requests_total{endpoint='/recommend',method='GET',status_code='200'} 42.0
cinematch_inference_latency_ms 124.5
cinematch_drift_detected 0.0
```

3. Data Models

3.1 MovieRecommendation

Field	Type	Description
title	string	Movie title (original casing from TMDB dataset)
poster	string (URL)	TMDB poster URL (w500 size) or placeholder if unavailable
rating	float | string	IMDB vote average (0-10) or 'N/A'
overview	string	Movie plot summary

3.2 RecommendationResponse

Field	Type	Description
recommendations	array[MovieRecommendation]	Ordered list of up to 10 movies

4. ML Pipeline Internal Functions

Function	recommend(movie: str) -> list
Module	ml_pipeline/recommend.py
Input	Raw query string from user
Output	List of movie title strings (up to 10), or [] if not found
MLflow logs	query, search_type, matched_title, recommendations_count, response_time_sec

Function	load_data() -> pd.DataFrame
Module	ml_pipeline/data_loader.py
Input	None (reads from data/ directory)
Validates	Schema (9 required columns), missing values (<10%), duplicates, outliers
Output	pd.DataFrame — merged movies+credits (4800+ rows)
Side Effects	Saves data_validation_report.json to logs/

Function	preprocess(df, save_features) -> tuple
Module	ml_pipeline/preprocess.py
Input	df: pd.DataFrame, save_features: bool
Output	(df, baselines: dict, importance: dict)
Tag Weights	director(3x) + cast(2x) + genres(2x) + overview(1x) + keywords(1x)
Side Effects	Saves versioned features to ml_pipeline/feature_store/v_TIMESTAMP/

5. Prometheus Metrics Specification

Metric Name	Type	Labels	Description
cinematch_requests_total	Counter	method, endpoint, status_code	All API requests
cinematch_request_latency_seconds	Histogram	endpoint	Request latency (buckets: 0.05-5s)
cinematch_recommendations_total	Counter	status	Recommendation results
cinematch_tmdb_requests_total	Counter	status	TMDB API call results
cinematch_inference_latency_ms	Gauge	-	Last e2e inference time
cinematch_latency_violations_total	Counter	-	Times >200ms threshold
cinematch_active_requests	Gauge	-	Concurrent requests
cinematch_drift_detected	Gauge	-	1=drift detected, 0=normal
cinematch_drift_score	Gauge	feature	Per-feature drift score
cinematch_retraining_triggered_total	Counter	-	Auto-retraining count
cinematch_feedback_total	Counter	status	User feedback logged

6. Environment Variables

Variable	Required	Description
TMDB_API_KEY	Yes	TMDB v3 API key for poster/metadata fetching
MLFLOW_TRACKING_URI	No	MLflow tracking URI (defaults to http://mlflow:5000)
PYTHONPATH	Yes (Docker)	Set to /app so ml_pipeline imports resolve
REQUESTS_CA_BUNDLE	Yes (Docker)	SSL certificate bundle for HTTPS in Docker

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